

$$193) y = 2x \cdot e^{3x}$$

$$R: \left(-\frac{2}{3}; -\frac{4}{3e^2}\right)$$

$$194) y = e^{-x^2}$$

$$R: \left(\sqrt{\frac{1}{2}}; \frac{1}{\sqrt{e}}\right) \left(-\sqrt{\frac{1}{2}}; \frac{1}{\sqrt{e}}\right)$$

$$195) y = e^{-\frac{1}{2}x^2}$$

$$R: \left(1; \frac{1}{\sqrt{e}}\right) \left(-1; \frac{1}{\sqrt{e}}\right)$$

$$196) y = x \cdot e^{-x^2}$$

$$R: (0, 0) \left(\sqrt{\frac{3}{2}}; \frac{\sqrt{3}}{2e^3}\right) \left(-\sqrt{\frac{3}{2}}; -\frac{\sqrt{3}}{2e^3}\right)$$

$$197) y = \frac{\ln x}{x}$$

$$R: \left(\sqrt{e^3}; \frac{3}{2\sqrt{e^3}}\right)$$

$$198) y = \frac{x}{\ln x}$$

$$R: \left(e^2; \frac{e^2}{2}\right)$$

$$199) y = \frac{\ln x}{x^2}$$

$$R: \left(\sqrt[6]{e}; \frac{\sqrt[3]{e^2}}{6 \cdot e}\right)$$

RESOLUCIÓN DE EJERCICIOS DE PUNTOS DE INFLEXIÓN

$$186) \quad y = x^4 - \frac{3}{2}x^2 + \frac{1}{2}$$

$$* \quad y' = 4x^3 - 3x$$

$$* \quad y'' = 12x^2 - 3$$

$$12x^2 - 3 = 0$$

$$12x^2 = 3$$

$$x^2 = \frac{3}{12} = \frac{1}{4}$$

$$x = \pm \sqrt{\frac{1}{4}} \quad \begin{cases} x_1 = \frac{1}{2} \\ x_2 = -\frac{1}{2} \end{cases}$$

$$y_{\left(\frac{1}{2}\right)} = \left(\frac{1}{2}\right)^4 - \frac{3}{2}\left(\frac{1}{2}\right)^2 + \frac{1}{2} = \frac{1}{16} - \frac{3}{8} + \frac{1}{2} = \frac{1-6+8}{16} = +\frac{3}{16}$$

$$y_{\left(-\frac{1}{2}\right)} = \left(-\frac{1}{2}\right)^4 - \frac{3}{2}\left(-\frac{1}{2}\right)^2 + \frac{1}{2} = \frac{1}{16} - \frac{3}{8} + \frac{1}{2} = \frac{3}{16}$$

$$\text{Ptos. de inflexión} \quad \left(\frac{1}{2}; \frac{3}{16}\right) \quad \left(-\frac{1}{2}; \frac{3}{16}\right)$$

$$187) \quad y' = 18x^2 - 16x + 11$$

$$y'' = 36x - 16$$

$$36x - 16 = 0$$

$$36x = 16 \rightarrow x = \frac{16}{36} \rightarrow x = \frac{4}{9}$$

$$Y_{\left(\frac{4}{9}\right)} = 6 \cdot \left(\frac{4}{9}\right)^3 - 8 \cdot \left(\frac{4}{9}\right)^2 + 11 \left(\frac{4}{9}\right) - 2.$$

$$Y_{\left(\frac{4}{9}\right)} = \frac{384}{729} - \frac{128}{81} + \frac{44}{9} - \frac{18}{9} = \frac{384 - 1152 + 3564 - 1458}{729} = \frac{1338}{729} = 1,84$$

$$P.I. \left(\frac{4}{9}; 1,84 \right)$$

$$188) Y' = \frac{0 \cdot (1+x^2) - 1 \cdot 2x}{(1+x^2)^2} = - \frac{2x}{(1+x^2)^2}$$

$$Y'' = - \frac{[(2)(1+x^2) - 2x \cdot 2 \cdot (1+x^2) \cdot 2x]}{(1+x^2)^3} = - \frac{2+2x^2-8x^2}{(1+x^2)^3} = - \frac{2-6x^2}{(1+x^2)^3}$$

$$2-6x^2=0$$

$$6x^2=2$$

$$x^2 = \frac{2}{6} = \frac{1}{3}$$

$$x = \pm \sqrt{\frac{1}{3}}$$

$$x = \begin{cases} x_1 = \sqrt{\frac{1}{3}} \\ x_2 = -\sqrt{\frac{1}{3}} \end{cases}$$

$$Y_{(\sqrt{\frac{1}{3}})} = \frac{1}{1 + (\sqrt{\frac{1}{3}})^2} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$Y_{(-\sqrt{\frac{1}{3}})} = \frac{1}{1 + (-\sqrt{\frac{1}{3}})^2} = \frac{1}{1 + \frac{1}{3}} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$P.I. \left(\sqrt{\frac{1}{3}}; \frac{3}{4} \right) \left(-\sqrt{\frac{1}{3}}; \frac{3}{4} \right)$$

$$189) Y' = \frac{0 \cdot (4 + 9x^2) - 2(18x)}{(4 + 9x^2)^2} = -\frac{36x}{(4 + 9x^2)^2}$$

$$Y'' = -\frac{36(4 + 9x^2)^{-2} - 36x \cdot 2(4 + 9x^2)^{-3} \cdot 18x}{(4 + 9x^2)^4} =$$

$$= -\frac{144 + 324x^2 - 1296x^2}{(4 + 9x^2)^3} = \frac{144 - 972x^2}{(4 + 9x^2)^3}$$

$$144 - 972x^2 = 0$$

$$972x^2 = 144$$

$$x^2 = \frac{144}{972} = \frac{4}{27} \rightarrow x = \pm \sqrt{\frac{4}{27}}$$

$$X = \begin{cases} x_1 = \frac{2}{\sqrt{27}} \\ x_2 = -\frac{2}{\sqrt{27}} \end{cases}$$

$$Y_{\left(\frac{4}{\sqrt{27}}\right)} = \frac{2}{4 + 9\left(\frac{4}{\sqrt{27}}\right)^2} = \frac{2}{4 + 9 \cdot \frac{16}{27}} = \frac{2}{4 + \frac{16}{3}} = \frac{2}{\frac{28}{3}} = \frac{6}{28} = \frac{3}{14}$$

$$Y_{\left(-\frac{4}{\sqrt{27}}\right)} = \frac{2}{4 + \left(-\frac{4}{\sqrt{27}}\right)^2} = \frac{2}{4 + \frac{16}{3}} = \frac{2}{\frac{28}{3}} = \frac{6}{28} = \frac{3}{14}$$

$$P.I. \left(\frac{4}{\sqrt{27}}; \frac{3}{14} \right) \left(-\frac{4}{\sqrt{27}}; \frac{3}{14} \right)$$

$$190) \quad Y' = \frac{1(1+x^2) - x(2x)}{(1+x^2)^2} = \frac{1+x^2-2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

$$Y'' = \frac{(-2x)(1+x^2)^{-2} - (1-x^2)2(1+x^2)^{-3} \cdot 2x}{(1+x^2)^4} =$$

$$= \frac{-2x - 2x^3 - 4x + 4x^3}{(1+x^2)^3} = \frac{-6x + 2x^3}{(1+x^2)^3}$$

$$-6x + 2x^3 = 0$$

$$2x(-3 + x^2) = 0$$

$$x_1 = 0$$

$$x^2 = 3 \rightarrow x = \pm\sqrt{3}$$

$$Y_{(0)} = \frac{0}{1+0} = 0$$

$$Y_{(\sqrt{3})} = \frac{\sqrt{3}}{1+(\sqrt{3})^2} = \frac{\sqrt{3}}{1+3} = \frac{\sqrt{3}}{4}$$

$$Y_{(-\sqrt{3})} = \frac{-\sqrt{3}}{1+(-\sqrt{3})^2} = \frac{-\sqrt{3}}{1+3} = -\frac{\sqrt{3}}{4}$$

$$\rightarrow \begin{cases} x_2 = \sqrt{3} \\ x_3 = -\sqrt{3} \end{cases}$$

$$P.I. (0, 0) \left(\sqrt{3}, \frac{\sqrt{3}}{4} \right) \left(-\sqrt{3}, -\frac{\sqrt{3}}{4} \right)$$

$$191) \quad y' = \frac{1(3x^2+1) - x \cdot (6x)}{(3x^2+1)^2} = \frac{3x^2+1-6x^2}{(3x^2+1)^2} = \frac{1-3x^2}{(3x^2+1)^2}$$

$$y'' = \frac{(-6x)(3x^2+1) - (1-3x^2) \cdot 6x}{(3x^2+1)^3} =$$

$$= \frac{-18x^3 - 6x - 12x + 36x^3}{(3x^2+1)^3} = \frac{18x^3 - 18x}{(3x^2+1)^3}$$

$$18x^3 - 18x = 0$$

$$18x(x^2-1) = 0$$

$$x^2 - 1 = 0$$

$$x = \pm \sqrt{1}$$

$$x_1 = 0.$$

$$x_2 = 1$$

$$x_3 = -1$$

$$y_{(0)} = \frac{0}{3 \cdot 0 + 1} = 0$$

$$y_{(1)} = \frac{1}{3 \cdot 1^2 + 1} = \frac{1}{4}$$

$$y_{(-1)} = \frac{-1}{3(-1)^2 + 1} = -\frac{1}{4}$$

$$P.I. \left(0, 0 \right) \left(1, \frac{1}{4} \right) \left(-1, -\frac{1}{4} \right)$$

$$192) y' = 1 \cdot e^{-x} + x \cdot (-e^{-x}) = 1e^{-x} - xe^{-x}$$

$$y'' = -e^{-x} - (1 \cdot e^{-x} - xe^{-x}) = -e^{-x} - e^{-x} + xe^{-x}$$

$$y'' = -2e^{-x} + x \cdot e^{-x}$$

$$-2e^{-x} + xe^{-x} = 0$$

$$e^{-x}(-2+x) = 0$$

$$-2+x = 0$$

$$x = 2$$

$$\boxed{x = 2}$$

$$y_{(2)} = 2 \cdot e^{-2} = \frac{2}{e^2}$$

$$P.I. \left(2 ; \frac{2}{e^2} \right)$$

$$193) \quad Y = 2x e^{3x}$$

$$Y' = 2e^{3x} + 2x \cdot 3e^{3x} = 2e^{3x} + 6xe^{3x}$$

$$Y'' = 2 \cdot 3e^{3x} + 6 \cdot e^{3x} + 6x \cdot 3e^{3x}$$

$$Y'' = \underline{6e^{3x} + 6e^{3x}} + 18x \cdot e^{3x} = 12e^{3x} + 18xe^{3x}$$

$$12e^{3x} + 18xe^{3x} = 0$$

$$6e^{3x}(2 + 3x) = 0$$

$$\begin{aligned} 2 + 3x &= 0 \\ 3x &= -2 \rightarrow \boxed{x = -\frac{2}{3}} \end{aligned}$$

$$Y\left(-\frac{2}{3}\right) = 2\left(-\frac{2}{3}\right)e^{3\left(-\frac{2}{3}\right)} = -\frac{4}{3}e^{-2} = -\frac{4}{3e^2}$$

$$P.I. \left(-\frac{2}{3}; -\frac{4}{3e^2}\right)$$

$$194) \quad Y = e^{-x^2}$$

$$Y' = -2x e^{-x^2}$$

$$Y'' = -\left[2e^{-x^2} + 2x \cdot (-2x)e^{-x^2}\right] = -\left[2e^{-x^2} - 4x^2 e^{-x^2}\right]$$

$$2e^{-x^2} - 4x^2 e^{-x^2} = 0$$

$$2e^{-x^2}(1 - 2x^2) = 0$$

$$1 - 2x^2 = 0$$

$$-2x^2 = -1 \rightarrow 2x^2 = 1 \rightarrow x^2 = \frac{1}{2} \rightarrow x = \pm \sqrt{\frac{1}{2}}$$

$$x_1 = \sqrt{\frac{1}{2}} \quad x_2 = -\sqrt{\frac{1}{2}}$$

$$Y\left(\sqrt{\frac{1}{2}}\right) = e^{-\left(\sqrt{\frac{1}{2}}\right)^2} = e^{-\frac{1}{2}} = \frac{1}{e^{\frac{1}{2}}} = \frac{1}{\sqrt{e}} \quad P.I. \left(\sqrt{\frac{1}{2}}; \frac{1}{\sqrt{e}}\right)$$

$$Y\left(-\sqrt{\frac{1}{2}}\right) = e^{-\left(-\sqrt{\frac{1}{2}}\right)^2} = e^{-\frac{1}{2}} = \frac{1}{e^{\frac{1}{2}}} = \frac{1}{\sqrt{e}} \quad P.I. \left(-\sqrt{\frac{1}{2}}; \frac{1}{\sqrt{e}}\right)$$

$$197) \quad y = \frac{\ln x}{x}$$

$$y' = \frac{\frac{1}{x} \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}$$

$$y'' = \frac{(0 - \frac{1}{x}) \cdot x^2 - (1 - \ln x) \cdot 2x}{x^4} = \frac{-x - 2x + 2x \ln x}{x^4} =$$

$$y'' = \frac{-3x + 2x \ln x}{x^4}$$

$$-3x + 2x \ln x = 0 \rightarrow 2x \ln x = 3x \rightarrow \ln x = \frac{3x}{2x}$$

$$\rightarrow \ln x = \frac{3}{2} \rightarrow x = e^{\frac{3}{2}} \rightarrow \boxed{x = \sqrt{e^3}}$$

$$y(\sqrt{e^3}) = \frac{\ln \sqrt{e^3}}{\sqrt{e^3}} = \frac{\ln e^{\frac{3}{2}}}{\sqrt{e^3}} = \frac{\frac{3}{2} \cdot \ln e}{\sqrt{e^3}} = \frac{\frac{3}{2} \cdot 1}{\sqrt{e^3}} = \frac{3}{2\sqrt{e^3}}$$

$$P.I. \left(\sqrt{e^3}; \frac{3}{2\sqrt{e^3}} \right)$$

$$198) \quad y = \frac{x}{\ln x}$$

$$y' = \frac{1 \cdot \ln x - x \cdot \frac{1}{x}}{(\ln x)^2} = \frac{\ln x - 1}{(\ln x)^2}$$

$$y'' = \frac{(\frac{1}{x} - 0)(\ln x)^2 - (\ln x - 1) \cdot 2(\ln x) \cdot (\frac{1}{x})}{(\ln x)^4} = \frac{\frac{1}{x} \ln x - 2 \frac{1}{x} \ln x + \frac{2}{x}}{(\ln x)^3}$$

$$y'' = \frac{-\frac{1}{x} \ln x + \frac{2}{x}}{(\ln x)^3} \rightarrow -\frac{1}{x} \ln x + \frac{2}{x} = 0 \rightarrow -\frac{1}{x} \ln x = -\frac{2}{x}$$

$$\rightarrow \frac{1}{x} \ln x = \frac{2}{x} \rightarrow \ln x = 2 \frac{x}{x} \rightarrow \ln x = 2 \rightarrow \boxed{x = e^2}$$

$$y(e^2) = \frac{e^2}{\ln e^2} = \frac{e^2}{2 \ln e} = \frac{e^2}{2} \quad P.I. \left(e^2; \frac{e^2}{2} \right)$$

$$199) \gamma = \frac{\ln x}{x^2}$$

$$\gamma' = \frac{\frac{1}{x} \cdot x^2 - (\ln x) \cdot 2x}{(x^4)} = \frac{x - 2x \ln x}{x^4}$$

$$\gamma'' = \frac{\left[1 - \left(2 \ln x + 2x \cdot \frac{1}{x}\right)\right] x^4 - (x - 2x \ln x) 4x^3}{x^8}$$

$$\gamma'' = \frac{\cancel{x^4} - 2x^4 \ln x + \cancel{2x^4} - \cancel{4x^4} + 8x^4 \ln x}{x^8}$$

$$\gamma'' = \frac{-x^4 + 6x^4 \ln x}{x^8} = \frac{x^4(-1 + 6 \ln x)}{x^8} = \frac{-1 + 6 \ln x}{x^2}$$

$$-1 + 6 \ln x = 0 \rightarrow 6 \ln x = 1 \rightarrow \ln x = \frac{1}{6} \rightarrow \boxed{x = e^{\frac{1}{6}}}$$

$$\gamma_{\left(e^{\frac{1}{6}}\right)} = \frac{\ln e^{\frac{1}{6}}}{\left(e^{\frac{1}{6}}\right)^2} = \frac{\frac{1}{6} \ln e}{e^{\frac{1}{3}}} = \frac{\frac{1}{6}}{\sqrt[3]{e}} = \frac{1}{6\sqrt[3]{e}}$$

$$\frac{1}{6\sqrt[3]{e}} \cdot \frac{\sqrt[3]{e^2}}{\sqrt[3]{e^2}} = \frac{\sqrt[3]{e^2}}{6\sqrt[3]{e \cdot e^2}} = \frac{\sqrt[3]{e^2}}{6\sqrt[3]{e^3}} = \frac{\sqrt[3]{e^2}}{6e}$$

$$P.I. \left(\sqrt[6]{e}, \frac{\sqrt[3]{e^2}}{6e} \right)$$